Bayesian A/B Testing on the Hillstrom Email Experiment

Semester V: Bayesian Inference
Indian Statistical Institute, Kolkata

Abstract

We analyse Kevin Hillstrom's MineThatData email experiment: 64,000 customers randomly assigned to a Men's email, a Women's email, or no email, with visits, conversions and spend recorded for two weeks. After checking the randomisation, we model visit and conversion rates per arm with Beta–Bernoulli posteriors and report the probability that an email beats no email, the probability that each arm is best, and the expected loss of choosing wrongly. We study how much the prior and the Bayes factor matter, using five priors and a subsampling experiment, and check that the credible intervals are calibrated. Revenue per customer is modelled with a hurdle model (Bernoulli for spending, lognormal with a Normal–Inverse-Gamma prior for the amount), and the fit is checked with posterior predictive checks. A hierarchical Beta–Binomial model, sampled with a hand-written Metropolis–Hastings sampler, estimates the email effect across customer-history segments with partial pooling. Everything is compared with standard frequentist tests, and a simulation of repeated looks at an A/A test shows that peeking inflates false positives, including for Bayesian stopping rules. All code uses only `numpy`, `pandas`, `scipy` and `matplotlib`, and every table and figure here is produced by `run_analysis.py` (seed 20260109, 20,000 posterior draws).

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1 Data and design

The data have 64,000 rows, one per customer who had bought in the previous twelve months. The columns used are the assigned arm (`segment`), the binary outcomes `visit` and `conversion`, the amount `spend`, and the covariates `history_segment`, `zip_code`, `channel`, `newbie`, `mens` and `womens`. We write "control" for no email, "Men's" and "Women's" for the two treatment arms.

Outcomes. `visit` is 1 if the customer visited the website in the following two weeks, `conversion` is 1 if the customer bought, and `spend` is the amount spent (0 for non-buyers). In this file the number of spenders equals the number of conversions in every arm, and no row has `spend` > 0 disagreeing with `conversion` $= 1$.

1.1 Randomisation checks

If assignment was random with equal probabilities, the arm counts should be close to equal thirds. The sample-ratio chi-square statistic is

$$\chi^2 = \sum_{k} \frac{(n_k - n/3)^2}{n/3} = 0.203 \quad (2 \text{ df}), \qquad p = 0.904.$$

Covariate balance was tested with a chi-square test of independence between arm and each covariate (Table 1). All p-values exceed 0.35 and Cramér's V is at most 0.011, so there is no sign of imbalance. These checks support, but cannot prove, that the randomisation was clean.

Table 1: Sample counts and covariate balance.

Arm	Count	Expected
Control	21,306	21,333.3
Men's	21,307	21,333.3
Women's	21,387	21,333.3

Variable	χ^2	df	p	Cramér's V
history_segment	13.153	12	0.358	0.0101
zip_code	2.872	4	0.579	0.0047
channel	3.632	4	0.458	0.0053
newbie	0.137	2	0.934	0.0015
mens	0.796	2	0.672	0.0035
womens	0.633	2	0.729	0.0031

2 Beta–Bernoulli analysis of visits and conversions

2.1 Posterior of a rate

For one arm let $y_i \in \{0, 1\}$, $s = \sum_i y_i$ and n the number of customers. The likelihood is

$$L(\theta) = \prod_{i=1}^{n} \theta^{y_i}(1-\theta)^{1-y_i} = \theta^s(1-\theta)^{n-s}.$$

With the prior $\pi(\theta) = \theta^{a-1}(1-\theta)^{b-1}/B(a,b)$, dropping everything that does not depend on θ gives

$$p(\theta \mid y) \;\propto\; \theta^s(1-\theta)^{n-s}\,\theta^{a-1}(1-\theta)^{b-1} = \theta^{(a+s)-1}(1-\theta)^{(b+n-s)-1}.$$

This is the kernel of a Beta density, so the normalising constant must be $1/B(a+s,\, b+n-s)$ and

$$\theta \mid y \sim \text{Beta}(a+s,\ b+n-s).$$

Unless stated otherwise we use $\text{Beta}(1,1)$.

2.2 Comparing two arms

The arms are independent a priori and the data are independent across arms, so the posteriors are independent, with densities f_A, f_B and CDFs F_A, F_B. Conditioning on $\theta_A = t$,

$$\mathbb{P}(\theta_B > \theta_A \mid \text{data}) = \int_0^1 f_A(t)\,\big(1 - F_B(t)\big)\,dt.$$

We estimate it by Monte Carlo: draw $\theta_A^{(m)} \sim f_A$ and $\theta_B^{(m)} \sim f_B$ independently for $m = 1, \ldots, M$ and use

$$\hat{P} = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\{\theta_B^{(m)} > \theta_A^{(m)}\}, \qquad \text{s.e.} = \sqrt{\hat{P}(1-\hat{P})/M}.$$

The same draws give the lift posteriors: absolute lift $\Delta^{(m)} = \theta_B^{(m)} - \theta_A^{(m)}$ and relative lift $\Delta^{(m)}/\theta_A^{(m)}$. Credible intervals are the 2.5% and 97.5% sample quantiles.

2.3 Probability of being best and expected loss

For K arms,

$$\mathbb{P}(\text{arm } k \text{ is best}) = \mathbb{P}(\theta_k > \theta_l \ \forall l \neq k) = \int f_k(t) \prod_{l \neq k} F_l(t)\,dt, \qquad \hat{P}_k = \frac{1}{M}\sum_m \mathbf{1}\{\arg\max_l \theta_l^{(m)} = k\},$$

and $\sum_k \hat{P}_k = 1$. The expected loss of committing to arm k is

$$L_k = \mathbb{E}\big[\max_l \theta_l - \theta_k\big] = \mathbb{E}[\max_l \theta_l] - \mathbb{E}[\theta_k], \qquad \hat{L}_k = \frac{1}{M}\sum_m \big(\max_l \theta_l^{(m)} - \theta_k^{(m)}\big).$$

$L_k \geq 0$ is the expected regret per customer of choosing arm k when the truth is drawn from the posterior. It is zero only if k is the best arm with probability one.

2.4 Results

Table 2 gives posteriors per arm and Table 3 the lifts against control.

Table 2: Posterior per arm under $\text{Beta}(1,1)$, with $\mathbb{P}$(best) and expected loss.

Outcome	Arm	n	s	Post. mean	95% CrI	P(best)	Exp. loss
Visit	Control	21,306	2,262	0.10621	[0.10211, 0.11037]	0.0000	7.66×10^{-2}
	Men's	21,307	3,894	0.18278	[0.17766, 0.18796]	1.0000	0
	Women's	21,387	3,238	0.15142	[0.14656, 0.15631]	0.0000	3.14×10^{-2}
Conversion	Control	21,306	122	0.00577	[0.00481, 0.00682]	0.0000	6.80×10^{-3}
	Men's	21,307	267	0.01258	[0.01110, 0.01410]	0.9999	6.84×10^{-9}
	Women's	21,387	189	0.00888	[0.00766, 0.01019]	0.0001	3.69×10^{-3}

Table 3: Lift against the no-email control.

Outcome	Arm	Abs. lift (95% CrI)	Rel. lift (95% CrI)	$\mathbb{P}(\theta_{\text{arm}} > \theta_{\text{ctrl}})$
Visit	Men's	0.0766 [0.0700, 0.0832]	0.722 [0.641, 0.806]	1.0000
	Women's	0.0452 [0.0389, 0.0516]	0.426 [0.356, 0.498]	1.0000
Conversion	Men's	0.0068 [0.0050, 0.0086]	1.196 [0.768, 1.702]	1.0000
	Women's	0.0031 [0.0015, 0.0047]	0.552 [0.236, 0.937]	0.9999

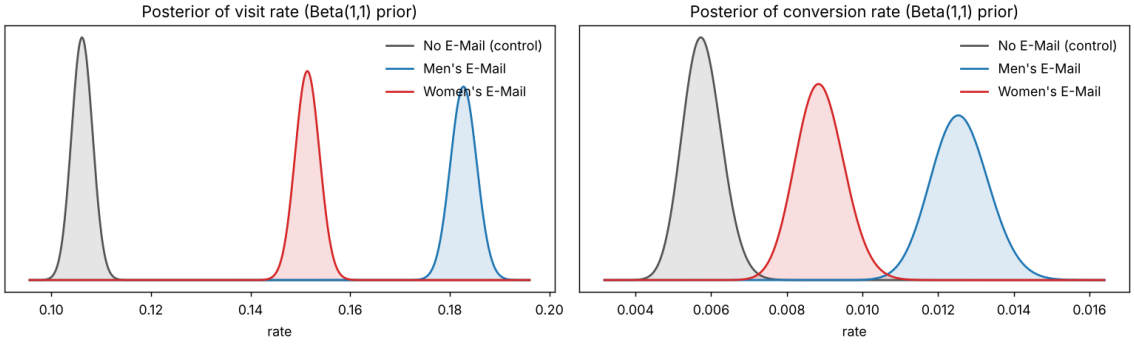


Figure 1: Posterior densities per arm for visit and conversion rates.

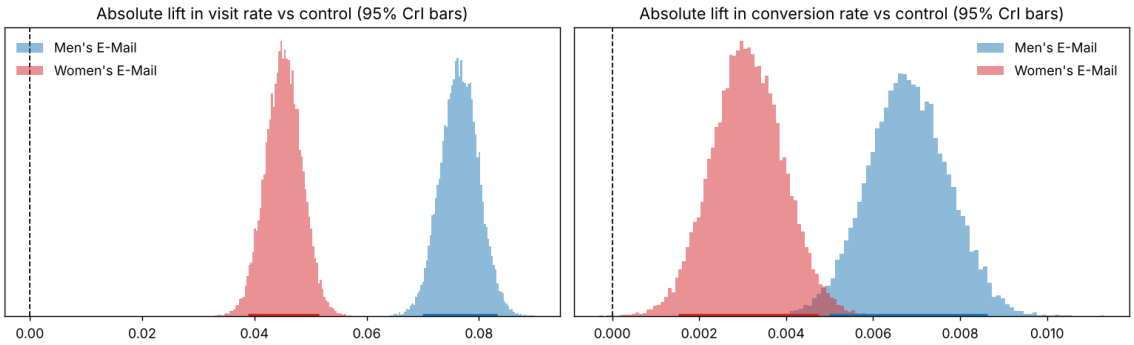


Figure 2: Posterior of the lift against control.

Both emails raise the visit rate and the conversion rate, and the Men's email is clearly the better of the two. It is best with probability 1.0000 for visits and 0.9999 for conversions. The Women's email still beats control for conversions with probability 0.9999, but its lift is smaller and its interval is wider, because there are only 122, 267 and 189 conversions in the three arms.

2.5 Sanity check with a normal approximation

For $\text{Beta}(a, b)$, $\text{Var} = ab/\big((a+b)^2(a+b+1)\big) \approx m(1-m)/(n+3)$ for large $a+b$, where m is the mean. For the control visit posterior $\text{Beta}(2263, 19045)$: $m = 0.10621$ and $a+b = 21308$, so $\text{Var} \approx 0.10621 \times 0.89379/21309 = 4.455 \times 10^{-6}$, against 4.441×10^{-6} from the Monte Carlo draws. The normal approximation to the comparison is

$$\mathbb{P}(\theta_B > \theta_A) \approx \Phi\left(\frac{m_B - m_A}{\sqrt{\text{Var}_A + \text{Var}_B}}\right).$$

For Women's versus control on conversion, $0.003115/\sqrt{2.693 \times 10^{-7} + 4.116 \times 10^{-7}} = 3.77$ and $\Phi(3.77) \approx 0.9999$, which matches the Monte Carlo value. The code prints this check for every arm.

3 Bayes factors and prior sensitivity

3.1 Beta–Binomial marginal likelihood and the Bayes factor

For one arm with prior $\mathrm{Beta}(a, b)$, the probability of a particular sequence with s successes in n trials, averaged over the prior, is

$$m(s, n) = \int_0^1 \theta^s (1-\theta)^{n-s} \frac{\theta^{a-1}(1-\theta)^{b-1}}{B(a, b)}\, d\theta = \frac{B(a+s,\, b+n-s)}{B(a, b)}.$$

Let H_0 say both arms share one rate (prior $\mathrm{Beta}(a, b)$) and H_1 say they have independent rates, each with a $\mathrm{Beta}(a, b)$ prior. The binomial coefficients are the same under both hypotheses and cancel, so

$$\mathrm{BF}_{01} = \frac{m(s_1 + s_2,\, n_1 + n_2)}{m(s_1, n_1)\, m(s_2, n_2)}.$$

We report $\log_{10} \mathrm{BF}_{01}$; negative values favour a difference between the arms.

3.2 Why the Bayes factor depends on the prior

A posterior forgets a weak prior as n grows, but a Bayes factor does not, because it compares prior predictive probabilities. A more diffuse prior under H_1 spreads mass over rates that the data rule out, which penalises H_1 (the Occam factor). So BF_{01} moves with the prior even when the posterior hardly does.

3.3 Results

Table 4 reports $\log_{10} \mathrm{BF}_{01}$ under five priors. The two pooled-rate priors are centred on the pooled rate of all three arms with strength 50 and 500, so they are data-informed and use the data twice.

Table 4: $\log_{10} \mathrm{BF}_{01}$ (negative favours an arm difference).

Outcome	Comparison	Beta(1, 1)	Jeffreys	Beta(0.1, 0.1)	Pooled, 50	Pooled, 500
Visit	Control vs Men's	−108.85	−108.82	−108.40	−109.51	−107.64
	Control vs Women's	−40.36	−40.34	−39.93	−41.17	−40.42
	Men's vs Women's	−14.35	−14.28	−13.83	−15.14	−15.04
Conversion	Control vs Men's	−9.46	−10.02	−10.07	−10.89	−11.19
	Control vs Women's	−0.43	−1.01	−1.08	−1.93	−2.37
	Men's vs Women's	−0.40	−0.89	−0.89	−1.73	−2.21

For visits the evidence for a difference is overwhelming under every prior. For conversions the Men's–control comparison is also decisive. The Women's–control comparison is the sensitive one: $\log_{10} \mathrm{BF}_{01}$ is -0.43 under $\mathrm{Beta}(1, 1)$ ($\mathrm{BF}_{10} \approx 2.7$, barely worth mentioning), -1.01 under Jeffreys, and -2.37 under the pooled prior of strength 500 ($\mathrm{BF}_{10} \approx 230$). Meanwhile the lift posterior is stable, with $\mathbb{P}(\text{Women's} > \text{control}) = 0.9999$. The posterior question ("how large is the lift?") and the Bayes-factor question ("is there any difference at all?") are different questions, and only the second is strongly prior-dependent.

3.4 Prior sensitivity by subsampling

We compared $\mathrm{Beta}(1, 1)$, Jeffreys $\mathrm{Beta}(0.5, 0.5)$ and a pooled-rate prior of strength 50 (data-informed, rebuilt from each subsample's own pooled rate). For $n = 250, 1000, 5000$ customers per arm we drew 100 subsamples without replacement and recorded the posterior-mean lift and $\mathbb{P}(\theta_{\text{arm}} > \theta_{\text{ctrl}})$, then repeated the analysis on the full data. Since outcomes are binary, the

number of successes in a without-replacement subsample of size n from an arm with S successes among N customers is exactly Hypergeometric$(S, N - S, n)$, so counts are drawn directly.

Table 5: Conversion, Men's vs control: posterior-mean lift and mean $\mathbb{P}$(Men's > control) across replicates.

	Beta$(1,1)$		Pooled, strength 50	
n per arm	lift (mean, sd)	P(Men's > ctrl)	lift (mean, sd)	P(Men's > ctrl)
250	0.00683 (0.00748)	0.718	0.00573 (0.00629)	0.739
1000	0.00623 (0.00398)	0.858	0.00594 (0.00380)	0.863
5000	0.00672 (0.00167)	0.997	0.00665 (0.00166)	0.997
Full	0.00680	1.000	0.00679	1.000

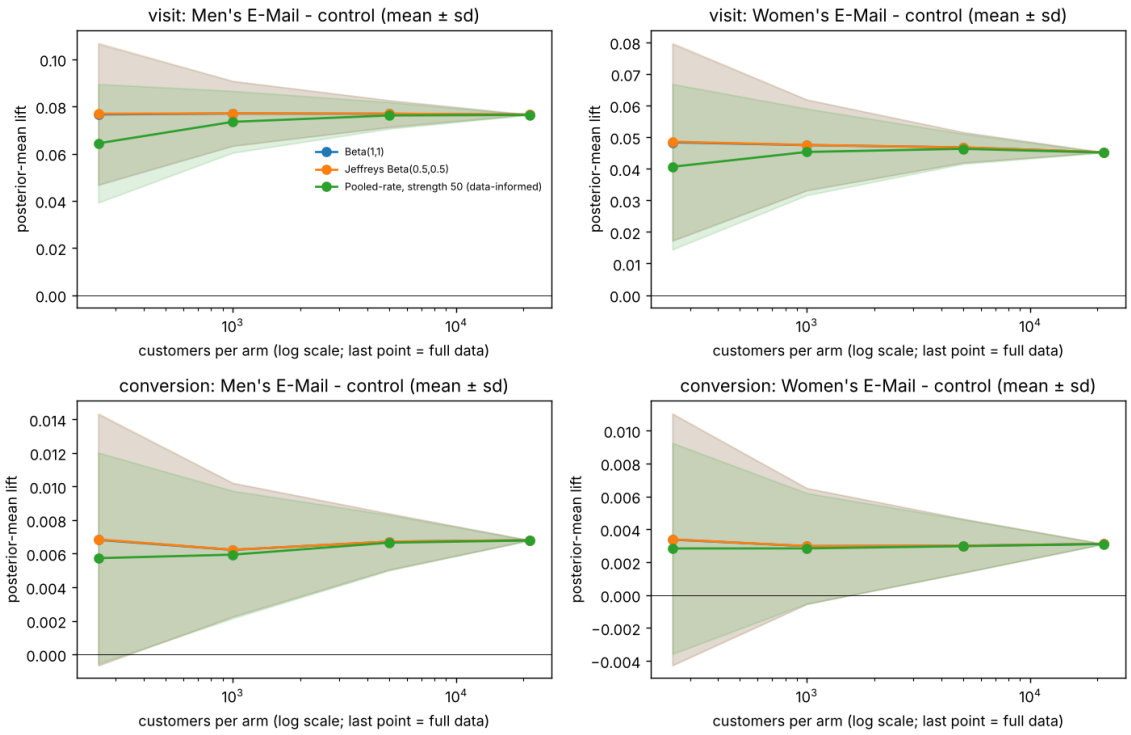


Figure 3: Prior sensitivity across subsample sizes.

On the full data the three priors' conversion lifts differ by at most 0.00002 in absolute terms, so the likelihood dominates. At $n = 250$ per arm the strength-50 pooled prior shrinks the mean lift by 16% (the same relative factor for visits and conversions), and the conversion posterior is mostly noise: $\mathbb{P}$(Men's > control) averages 0.72 for conversion against 0.96 for visits. So the prior matters for a rare outcome at small n, but not here with $\approx$ 21,000 customers per arm.

4 Calibration of credible intervals

If θ is drawn from the prior used for fitting, $s \sim \text{Bin}(n, \theta)$, and a 95% equal-tailed interval is built from the posterior, then the interval contains θ with probability exactly 0.95 on average over this process. We checked this by simulation (20,000 replications per row, Monte Carlo s.e. about 0.0015) and compared with a deliberately mismatched case, where data come from a Beta$(1, 1)$ truth but the pooled-rate prior of strength 50 is used for fitting.

Table 6: Coverage of 95% credible intervals.

Fit prior	Truth drawn from	n	Coverage	Mean width
Beta$(1,1)$	same	250 / 1000 / 5000 / 21000	0.952 / 0.952 / 0.953 / 0.953	0.097 → 0.011
Jeffreys	same	250 / 1000 / 5000 / 21000	0.948 / 0.949 / 0.952 / 0.952	0.079 → 0.009
Pooled, 50	same	250 / 1000 / 5000 / 21000	0.951 / 0.949 / 0.951 / 0.950	0.017 → 0.002
Pooled, 50	Beta$(1,1)$ (mismatch)	250 / 1000 / 5000 / 21000	0.308 / 0.562 / 0.790 / 0.897	0.095 → 0.011

With matched priors, coverage is between 0.948 and 0.953 at every n, as it should be. The mismatched row shows what the guarantee does not cover: a strong prior concentrated in the wrong place gives intervals that are too narrow and miss the truth, and only a large n repairs this.

5 Revenue per customer: a hurdle model

Spend is zero for about 99% of customers, and positive amounts are right-skewed. We model it in two parts. Let $c_i = \mathbf{1}\{\text{spend}_i > 0\}$, with $c_i \sim \text{Bernoulli}(\theta)$, and for spenders let $\log y_i \mid \mu, \sigma^2 \sim \mathcal{N}(\mu, \sigma^2)$.

5.1 The posterior factorises

Let $n_1 = \sum_i c_i$. The likelihood is

$$\prod_i \theta^{c_i}(1-\theta)^{1-c_i} \times \prod_{i:c_i=1} f(y_i \mid \mu, \sigma^2) = \big[\theta^{n_1}(1-\theta)^{n-n_1}\big] \times \Big[\prod_{c_i=1} f(y_i \mid \mu, \sigma^2)\Big].$$

The prior factorises as $\pi(\theta)\,\pi(\mu, \sigma^2)$, so the posterior is a product of a function of θ alone and a function of (μ, σ^2) alone. They are independent a posteriori and are sampled separately:

$$\theta \mid \text{data} \sim \text{Beta}(1+n_1,\ 1+n-n_1), \qquad (\mu, \sigma^2) \mid \text{data} \sim \text{NIG}(k_n, m_n, a_n, b_n),$$

where the second posterior uses only the n_1 positive amounts.

5.2 Normal–Inverse-Gamma posterior

Let $x_i = \log y_i$ for the spenders, $x_i \mid \mu, \sigma^2 \sim \mathcal{N}(\mu, \sigma^2)$, with the prior $\mu \mid \sigma^2 \sim \mathcal{N}(m_0, \sigma^2/k_0)$ and $\sigma^2 \sim \text{InvGamma}(a_0, b_0)$. Write $S = \sum_i (x_i - \bar{x})^2$, so that $\sum_i (x_i - \mu)^2 = S + n(\bar{x} - \mu)^2$. Then

$$\begin{aligned} \text{likelihood} &\propto (\sigma^2)^{-n/2} \exp\Big\{-\tfrac{S + n(\bar{x}-\mu)^2}{2\sigma^2}\Big\}, \\ \text{prior} &\propto (\sigma^2)^{-1/2} \exp\Big\{-\tfrac{k_0(\mu - m_0)^2}{2\sigma^2}\Big\}\,(\sigma^2)^{-(a_0+1)} \exp\Big\{-\tfrac{b_0}{\sigma^2}\Big\}. \end{aligned}$$

Completing the square. With $k_n = k_0 + n$ and $m_n = (k_0 m_0 + n\bar{x})/k_n$,

$$\begin{aligned} n(\bar{x}-\mu)^2 + k_0(\mu - m_0)^2 &= k_n\mu^2 - 2\mu(n\bar{x} + k_0 m_0) + (n\bar{x}^2 + k_0 m_0^2) \\ &= k_n(\mu - m_n)^2 - k_n m_n^2 + n\bar{x}^2 + k_0 m_0^2. \end{aligned}$$

The remaining constant is

$$\begin{aligned} n\bar{x}^2 + k_0 m_0^2 - \frac{(n\bar{x} + k_0 m_0)^2}{k_n} &= \frac{(n+k_0)(n\bar{x}^2 + k_0 m_0^2) - (n\bar{x} + k_0 m_0)^2}{k_n} \\ &= \frac{nk_0\bar{x}^2 + nk_0 m_0^2 - 2nk_0\bar{x}m_0}{k_n} = \frac{nk_0}{k_n}(\bar{x} - m_0)^2. \end{aligned}$$

Hence

$$n(\bar{x}-\mu)^2 + k_0(\mu - m_0)^2 = k_n(\mu - m_n)^2 + \frac{k_0 n}{k_n}(\bar{x} - m_0)^2.$$

Collecting powers of σ^2. Multiplying likelihood and prior, the exponent of σ^2 is $-n/2 - 1/2 - (a_0 + 1)$. Split it as $-1/2$ (belongs to the Normal kernel for μ) and $-(a_0 + n/2 + 1)$ (the Inverse-Gamma kernel). The exponential term becomes $\exp\{-[2b_0 + S + \frac{k_0 n}{k_n}(\bar{x} - m_0)^2 + k_n(\mu - m_n)^2]/(2\sigma^2)\}$, so

$$p(\mu, \sigma^2 \mid x) \propto (\sigma^2)^{-1/2} \exp\Big\{-\tfrac{k_n(\mu - m_n)^2}{2\sigma^2}\Big\} \cdot (\sigma^2)^{-(a_n+1)} \exp\Big\{-\tfrac{b_n}{\sigma^2}\Big\},$$

$$a_n = a_0 + \frac{n}{2}, \qquad b_n = b_0 + \frac{S}{2} + \frac{k_0 n(\bar{x} - m_0)^2}{2k_n}.$$

That is, $\sigma^2 \mid x \sim \text{InvGamma}(a_n, b_n)$ and $\mu \mid \sigma^2, x \sim \mathcal{N}(m_n, \sigma^2/k_n)$. We sample $\sigma^2 = 1/\text{Gamma}(a_n, \text{rate} = b_n)$ and then μ. The code checks this against a brute-force grid posterior.

Prior values. $\theta \sim \text{Beta}(1, 1)$ and $k_0 = 1$, $a_0 = 2$, $m_0 = 4.3952$ (pooled mean log spend among spenders), $b_0 = 0.7028$ (pooled variance of log spend times $a_0 - 1$). These hyperparameters are centred on the data (weakly informative by design, but data-informed).

5.3 Expected revenue

$\mathbb{E}[Y] = \mathbb{P}(Y > 0)\,\mathbb{E}[Y \mid Y > 0] = \theta\,\mathbb{E}[e^X]$ with $X \sim \mathcal{N}(\mu, \sigma^2)$. The Normal moment generating function at $t = 1$ gives $\mathbb{E}[e^X] = \exp(\mu + \sigma^2/2)$, so

$$R(\theta, \mu, \sigma^2) = \theta\, \exp(\mu + \sigma^2/2).$$

The posterior of R is obtained by applying this function to each joint draw $(\theta^{(m)}, \mu^{(m)}, \sigma^{2(m)})$.

5.4 Results

Table 7: Expected revenue per customer ($) under the hurdle model.

Arm	Customers	Spenders	Observed mean	Post. mean	95% CrI	P(best)	Exp. loss
Control	21,306	122	0.653	0.653	[0.511, 0.824]	0.0000	0.7457
Men's	21,307	267	1.423	1.397	[1.175, 1.643]	0.9691	0.0020
Women's	21,387	189	1.077	1.091	[0.893, 1.327]	0.0309	0.3078

Table 8: Incremental revenue per customer against control ($).

Arm	Mean	95% CrI	$\mathbb{P}(> 0)$
Men's	0.7437	[0.4612, 1.0294]	1.0000
Women's	0.4379	[0.1737, 0.7128]	0.9994

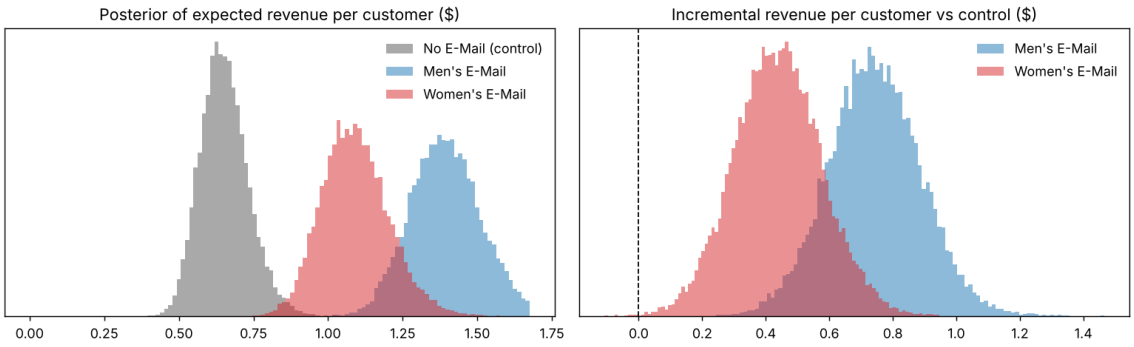


Figure 4: Posterior of expected revenue per customer by arm.

The Men's email earns about \$0.74 more per customer than no email, and the Women's email about \$0.44 more. The Men's email is the best arm for revenue with probability 0.969.

5.5 Posterior predictive checks

For each of 2,000 replications we drew (μ, σ^2) from the posterior, simulated n_1 lognormal amounts, and computed the mean, 90th percentile and maximum. The Bayesian p-value is $\mathbb{P}(T_{\text{rep}} \geq T_{\text{obs}})$; values near 0 or 1 signal misfit.

Table 9: Posterior predictive checks on positive spend.

Arm	Statistic	Observed	Replicated mean	Replicated 95% range	Bayesian p
Control	mean	114.0	113.6	[90.1, 145.2]	0.443
	p90	226.9	226.4	[170.0, 310.0]	0.443
	max	499.0	694.5	[347.6, 1518.9]	0.732
Men's	mean	113.5	111.1	[93.9, 133.0]	0.367
	p90	262.8	230.1	[183.6, 290.8]	0.112
	max	499.0	923.9	[496.5, 1878.7]	0.973
Women's	mean	121.9	122.8	[101.2, 148.5]	0.514
	p90	268.0	251.1	[193.8, 320.2]	0.276
	max	499.0	888.7	[445.0, 1879.4]	0.946

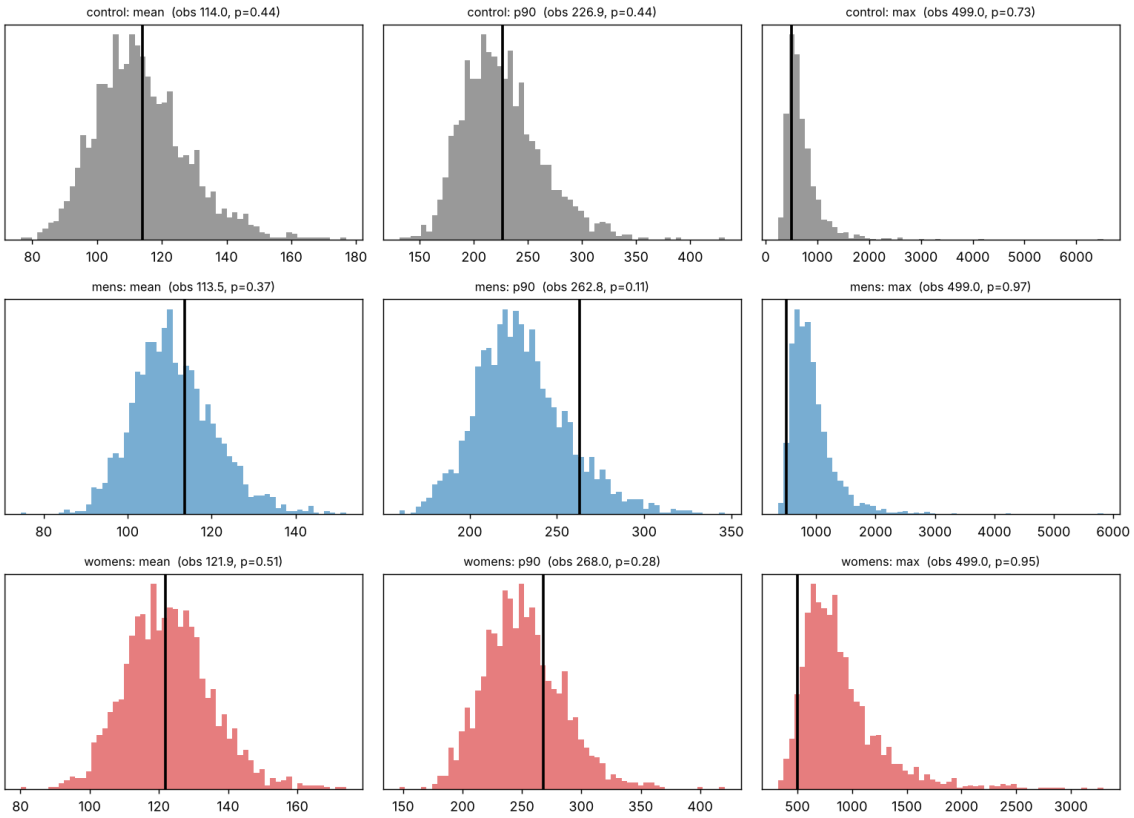


Figure 5: Posterior predictive checks for positive spend.

The lognormal reproduces the mean and the 90th percentile (Bayesian p between 0.11 and 0.51). It overpredicts the maximum ($p = 0.73$, 0.97, 0.95), and the model puts 1.2% to 1.8% of positive amounts above \$499 while none are observed. In the data, 12 spenders (2, 6 and 4 by arm) sit exactly at \$499, so spend is capped there, and a lognormal cannot reproduce that. The upper tail is therefore too heavy, and tail-driven quantities should be read as approximate. The comparison of means between arms is less affected, because the mean and p90 fit well.

6 Hierarchical model across customer-history segments

Conversions were split by `history_segment` (seven spend-history bands). Segments with few customers give noisy rates, so we let them borrow strength from each other. This is done separately for each arm.

6.1 Model

For segment g, with s_g conversions among n_g customers,

$$s_g \mid \theta_g \sim \mathrm{Bin}(n_g, \theta_g), \qquad \theta_g \mid \mu, \kappa \sim \mathrm{Beta}(\alpha, \beta), \quad \alpha = \mu\kappa,\ \beta = (1-\mu)\kappa.$$

Here μ is the common mean rate and κ controls how tightly the segment rates cluster around it (large κ means strong pooling).

Integrating out θ_g.

$$p(s_g \mid \mu, \kappa) = \binom{n_g}{s_g} \int_0^1 \frac{\theta^{s_g+\alpha-1}(1-\theta)^{n_g-s_g+\beta-1}}{B(\alpha,\beta)}\, d\theta = \binom{n_g}{s_g} \frac{B(\alpha+s_g, \beta+n_g-s_g)}{B(\alpha,\beta)}.$$

Exact conditional of θ_g. By the same kernel argument as in Section 2,

$$\theta_g \mid \mu, \kappa, s_g \sim \text{Beta}(\mu\kappa + s_g,\ (1-\mu)\kappa + n_g - s_g).$$

So the sampler only has to explore (μ, κ); the θ_g are then drawn exactly.

6.2 Metropolis–Hastings sampler

Let $\eta = (\eta_1, \eta_2) = (\text{logit}\,\mu,\ \log\kappa)$ with priors

$$\eta_1 \sim \mathcal{N}\big(\text{logit(pooled rate)},\, 2^2\big), \qquad \eta_2 \sim \mathcal{N}\big(\log 100,\, 1.5^2\big).$$

Then, up to a constant,

$$p(\eta \mid \text{data}) \propto \prod_g \frac{B(\alpha + s_g, \beta + n_g - s_g)}{B(\alpha, \beta)}\, \mathcal{N}(\eta_1; \cdot)\, \mathcal{N}(\eta_2; \cdot).$$

The proposal is a random walk $\eta^* = \eta + e$ with $e \sim \mathcal{N}(0, \Sigma)$. It is symmetric, $q(\eta^* \mid \eta) = q(\eta \mid \eta^*)$, so the Hastings ratio reduces to

$$r = \frac{p(\eta^* \mid \text{data})}{p(\eta \mid \text{data})}, \qquad \text{accept with probability } \min(1, r).$$

Because the priors are written directly as densities in η, there is no Jacobian. (Had they been placed on (μ, κ), r would pick up the factor $\mu^*(1-\mu^*)\kappa^*/(\mu(1-\mu)\kappa)$.) The step size and the coordinate scales are adapted during burn-in only and then frozen, so the retained draws come from a fixed symmetric kernel. We ran 4 chains from over-dispersed starting points, 4,000 iterations each with 1,000 burn-in, giving 12,000 retained draws per arm.

6.3 Convergence

Table 10: Sampler diagnostics.

Arm	Parameter	Post. mean	Post. sd	Split $\hat{R}$	ESS (of 12,000)	Acceptance (4 chains)
Control	logit μ	−4.952	0.244	1.004	1282	0.34, 0.29, 0.23, 0.27
	log κ	6.390	0.813	1.004	1267	
Men's	logit μ	−4.126	0.201	1.004	1230	0.23, 0.29, 0.30, 0.28
	log κ	5.853	0.727	1.003	1210	
Women's	logit μ	−4.448	0.228	1.004	1167	0.30, 0.38, 0.29, 0.36
	log κ	5.982	0.811	1.002	1151	

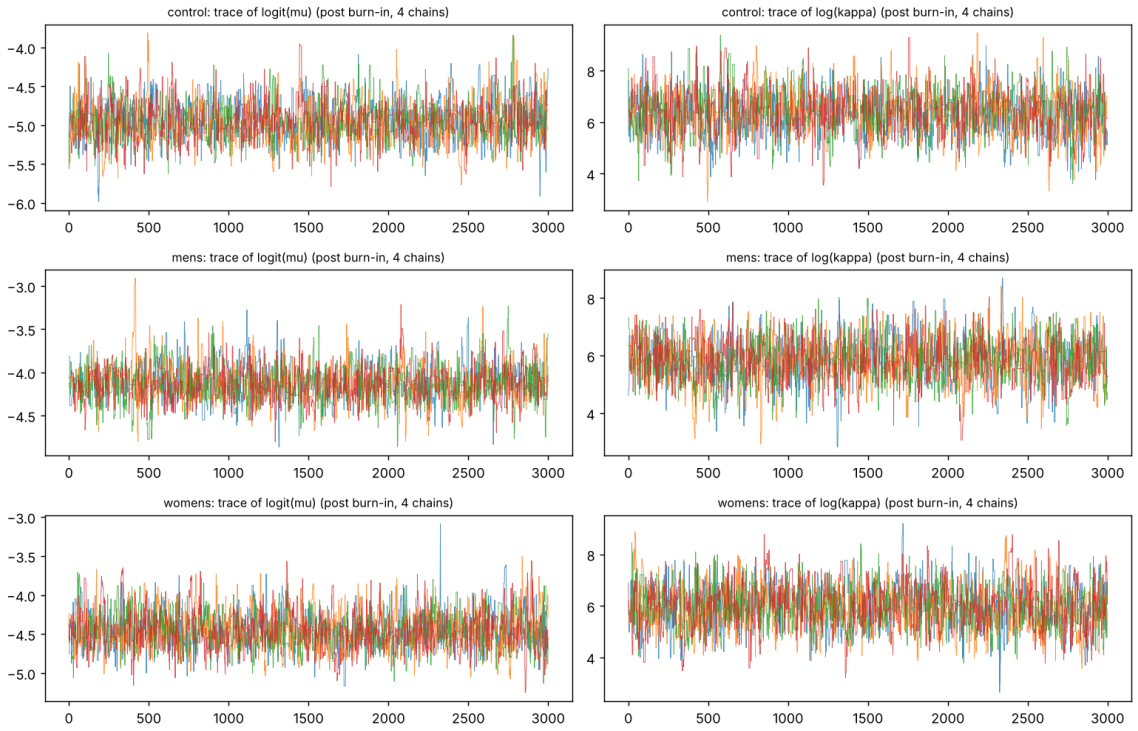


Figure 6: Trace plots of the hand-written MH sampler.

The split $\hat{R}$ is at most 1.004, the effective sample size is above 1,150 for every parameter, and acceptance rates are between 0.23 and 0.38.

6.4 Segment-level lifts

Table 11: Conversions per segment and arm (s/n).

Segment	Control	Men's	Women's
1) \$0–\$100	32/7612	71/7724	61/7634
2) \$100–\$200	20/4836	42/4691	37/4727
3) \$200–\$350	28/4044	63/4090	26/4155
4) \$350–\$500	24/2124	42/2097	20/2188
5) \$500–\$750	9/1652	22/1597	26/1662
6) \$750–\$1,000	3/622	16/644	8/593
7) \$1,000+	6/416	11/464	11/428

Table 12: Segment-level conversion lift: no pooling vs partial pooling. The last column is the CrI width ratio (partial/no pooling).

Comparison	Segment	No pool: lift	No pool: 95% CrI	Partial: lift	Partial: 95% CrI	Ratio
Men's − control	1) \$0–\$100	0.00498	[0.00240, 0.00762]	0.00506	[0.00252, 0.00772]	1.00
	2) \$100–\$200	0.00483	[0.00156, 0.00828]	0.00502	[0.00184, 0.00841]	0.98
	3) \$200–\$350	0.00848	[0.00390, 0.01324]	0.00851	[0.00435, 0.01292]	0.92
	4) \$350–\$500	0.00871	[0.00118, 0.01640]	0.00910	[0.00233, 0.01612]	0.91
	5) \$500–\$750	0.00829	[0.00163, 0.01541]	0.00829	[0.00230, 0.01473]	0.90
	6) \$750–\$1,000	0.01991	[0.00672, 0.03470]	0.01553	[0.00577, 0.02719]	0.77
	7) \$1,000+	0.00907	[−0.00963, 0.02861]	0.01003	[−0.00257, 0.02348]	0.68
Women's − control	1) \$0–\$100	0.00379	[0.00128, 0.00633]	0.00372	[0.00134, 0.00619]	0.96
	2) \$100–\$200	0.00370	[0.00062, 0.00688]	0.00365	[0.00065, 0.00677]	0.98
	3) \$200–\$350	−0.00066	[−0.00425, 0.00296]	−0.00010	[−0.00357, 0.00330]	0.95
	4) \$350–\$500	−0.00222	[−0.00857, 0.00377]	−0.00070	[−0.00640, 0.00475]	0.90
	5) \$500–\$750	0.01020	[0.00327, 0.01768]	0.00881	[0.00279, 0.01537]	0.87
	6) \$750–\$1,000	0.00864	[−0.00236, 0.02106]	0.00667	[−0.00122, 0.01599]	0.73
	7) \$1,000+	0.01117	[−0.00819, 0.03164]	0.00858	[−0.00435, 0.02322]	0.69

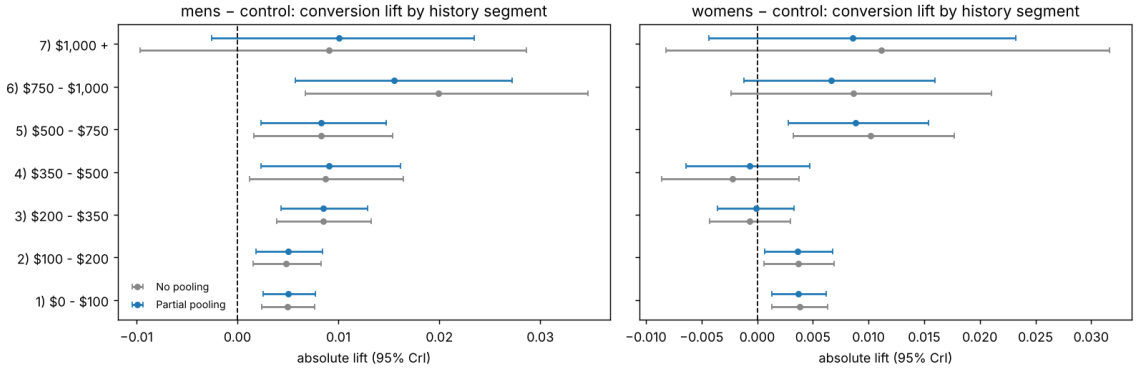


Figure 7: Segment-level lifts, no pooling versus partial pooling.

Pooling is mild because the segments are fairly homogeneous, but it helps where data are thinnest. In the two highest-spend segments the credible intervals shrink to 68%–77% of their no-pooling width, while in the largest segment they barely change (ratio 1.00 for Men's, 0.96 for Women's). For example, the Men's–control lift in the \$750–\$1,000 segment moves from 0.0199 (no pooling) to 0.0155 (partial pooling), pulled towards the other segments. Seven segments with few conversions each and no multiplicity correction means these segment-level differences are exploratory.

7 Comparison with frequentist methods

For binary outcomes we used a two-proportion z-test with a Wald interval, and for spend a Welch t-test and a percentile bootstrap (2,000 resamples).

Table 13: Frequentist and Bayesian results on the same data.

Outcome	Comparison	Observed diff.	Frequentist 95% CI	Bayes mean diff.	Bayes 95% CrI
Visit	Men's − ctrl	0.07659	[0.06995, 0.08323]	0.07657	[0.07000, 0.08318]
Visit	Women's − ctrl	0.04523	[0.03889, 0.05157]	0.04522	[0.03888, 0.05155]
Conversion	Men's − ctrl	0.00681	[0.00500, 0.00861]	0.00680	[0.00500, 0.00862]
Conversion	Women's − ctrl	0.00311	[0.00150, 0.00472]	0.00311	[0.00152, 0.00474]
Spend ($)	Men's − ctrl	0.7698	Welch [0.485, 1.055]	0.7437	[0.4612, 1.0294]
Spend ($)	Women's − ctrl	0.4244	Welch [0.169, 0.680]	0.4379	[0.1737, 0.7128]

The z-test p-values are below 10^{-16} (visit), 1.5×10^{-13} (Men's conversion) and 1.6×10^{-4} (Women's conversion); the Welch p-values for spend are 1.2×10^{-7} and 1.1×10^{-3}. Bootstrap intervals for spend ([0.483, 1.070] and [0.169, 0.668]) are nearly the same width as the Welch intervals. With this much data and a flat prior, Bayesian and frequentist intervals are almost identical. The Bayesian version adds direct statements such as $\mathbb{P}(\text{arm} > \text{control})$, $\mathbb{P}(\text{best})$ and expected loss. The spend row for the Bayesian method is the hurdle-model incremental expected revenue, a model-based estimand, so it need not equal the raw difference in means.

8 Peeking in an A/A test

We simulated A/A tests (true conversion rate 0.01 in both arms), with 2,000 customers per arm per batch, 10 looks and 2,000 simulated experiments, using Beta$(1, 1)$ priors. Three rules were applied at every look:

1. the naive two-sided z-test at the 5% level;
2. the posterior rule: flag if $\mathbb{P}(B > A) > 0.95$ or < 0.05;
3. the expected-loss rule: stop when $\min\big(\mathbb{E}[(A - B)_+], \mathbb{E}[(B - A)_+]\big) < \varepsilon$ with $\varepsilon = 0.0003$.

Table 14: Fraction of A/A experiments flagged (Monte Carlo s.e. near 0.05 is about 0.005).

Rule	Flagged at final look only	Flagged at any look
1. Naive z-test at 0.05	0.039	0.189
2. $\mathbb{P}(B > A) > 0.95$ or < 0.05	0.088	0.336
3. Expected loss $< \varepsilon$ (stop)	0.827	0.996

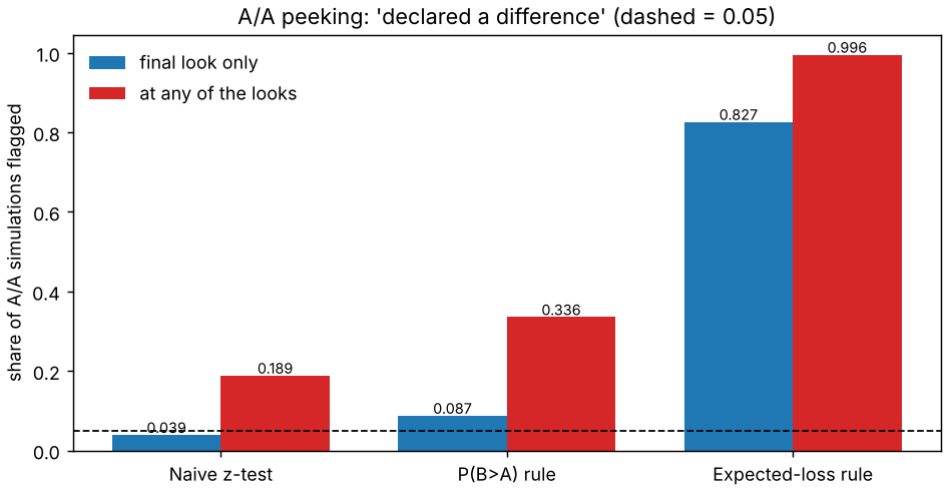


Figure 8: Peeking simulation: A/A false-positive rates.

The naive test is at about 4% at the single final look but flags 18.9% of A/A experiments if one stops at the first significant look. The posterior rule is worse: it has two 5% tails, so even a single look flags roughly 10% of A/A runs, and repeated looks push it to 33.6%. So Bayesian posterior-probability stopping is not immune to peeking in frequentist error terms.

Rule 3 stops in 99.6% of A/A runs, on average at look 2.7. This is the rule working as designed, because it bounds regret, not the false-positive rate. In an A/A test the true regret of any decision is exactly zero, so "stopping" there is not a false positive in the frequentist sense. In a supplementary A/B scenario (true rates 0.010 and 0.012, 1,000 simulations) the rule stops in 99.8% of runs at look 2.25 on average, ships B in 87.6% of runs, and has mean realised regret 2.5×10^{-4} per customer. The expected-loss rule should therefore not be described as controlling Type I error.

9 Limitations

- Outcomes were recorded for only two weeks, so nothing here speaks to long-term effects, repeat purchases, unsubscribes or fatigue.
- Spend is capped near \$499 and piles up there, so the lognormal tail is approximate.
- Segment-level effects are exploratory: seven segments, several with a handful of conversions, no multiplicity correction.
- Conclusions are causal only because assignment was randomised; the balance and sample-ratio checks support but do not prove a clean randomisation.
- The results describe one retailer's customers who had purchased in the last twelve months in 2008.
- The pooled-rate Beta priors and the data-centred Normal–Inverse-Gamma hyperparameters use the data twice, which is why they are labelled data-informed.
- Bayes factors against a "no difference" model are prior-sensitive (Section 3).

10 Conclusion

Both emails increase visits, conversions and revenue per customer, and the Men's email is better than the Women's email on all three. The Men's email raises conversion from 0.58% to 1.26% and expected revenue by about \$0.74 per customer; the Women's email raises conversion to 0.89% and revenue by about \$0.44. The Bayesian and frequentist answers agree closely at this sample size. The prior matters mainly for rare outcomes at small n and, most strongly, for Bayes factors on the weakly supported Women's-versus-control conversion comparison. The hierarchical model shows mild shrinkage across segments, and the simulation shows that sequential stopping rules, Bayesian or not, need to be judged by the error they are meant to control.

Data. Kevin Hillstrom, MineThatData E-Mail Analytics and Data Mining Challenge (2008), `http://blog.minethatdata.com/2008/03/minethatdata-e-mail-analytics-and-data.html`.